

WooCommerce + LMS Case Study

Case Study: WooCommerce + LMS Hosting Estimate

Use this case study after the hosting comparison checklist from the lesson. The goal is to turn a real plugin stack into a hosting shortlist with numbers, source checks, and a launch baseline.

Project Snapshot

Item	Assumption
Site type	WooCommerce plus learning management system (LMS)
Business model	Sell paid courses through WooCommerce
LMS model	Logged-in students watch lessons, complete quizzes, track progress, download materials
Launch size	20 courses, 1,000 registered students, 50 to 100 same-time active students during campaigns
Admin activity	Course editing, order management, refunds, reports, student support
Media model	Videos hosted outside WordPress, for example Vimeo, YouTube, Bunny, or S3-style storage
Local media	Images, PDFs, thumbnails, certificates, downloadable files
Risk level	High. Checkout, logins, course access, emails, and student progress must work reliably.

Realistic Plugin Stack

Layer	Example plugin or service	Hosting impact
Commerce	WooCommerce	Cart, checkout, orders, emails, webhooks, admin reports
LMS	LearnDash	Logged-in lessons, quizzes, progress tracking, database writes
Security	Defender	Malware scans, firewall, login protection (2FA), background load
SEO	SmartCrawl	Admin analysis, schema, sitemaps, redirects, indexing data
Cache and optimization	Hummingbird, LiteSpeed Cache, WP Rocket, or platform cache	Public-page cache, cache exclusions for cart, account, and lessons

Layer	Example plugin or service	Hosting impact
Email delivery	WP Mail SMTP or transactional email service	Order emails, enrollment emails, password resets, course notifications
Image optimization	Smush, Imagify, ShortPixel, or platform image service	Image processing, background tasks, media storage
Forms, CRM, analytics	Form plugin, CRM integration, analytics plugin	External API calls, cron jobs, database writes
Backups and staging	Host backup and staging tools	Storage use, restore process, safer updates

Requirements to Check

Always check the official requirements before choosing a plan:

- WooCommerce server recommendations: check for WordPress version, PHP version, database version, HTTPS support, and memory requirement
- LearnDash requirements: check for PHP memory, database, and server settings for the LMS workload
- Defender requirements: check for WordPress/PHP version, scan behavior, and background CPU impact
- SmartCrawl requirements: PHP modules and compatibility checks
- Cache plugin docs (Hummingbird, LiteSpeed Cache, WP Rocket): depends on your host stack
- WP Mail SMTP: PHP version requirements
- Image optimization plugins: storage, processing, and API limits

Sizing Calculation

Step 1: Minimum requirements

Use the highest requirement across the stack:

Area	Result
WordPress and WooCommerce versions	Use current supported versions
PHP	Use the highest PHP version your stack supports
Database	Meet WooCommerce and LMS database requirements
HTTPS	Required
PHP memory	Start at 512 MB minimum for this stack, with an option for 768 MB or 1 GB if imports, reports, page builder work, course admin, or heavy extensions hit memory limits

Step 2: Estimate dynamic workload

For this site, public marketing pages can be cached. The important load is dynamic:

Dynamic action	Why it matters
Checkout	Cart, payment, order creation, emails, webhooks
Student login	Sessions, account checks, redirects
Lesson view	Logged-in access checks, progress data
Quiz submission	Database writes, scoring
Course progress update	Database writes for each student action
Admin reports	Expensive queries, order and student data

Dynamic action	Why it matters
Security scans	Background file and database checks
Email sending	Order, enrollment, password, notification emails

Step 3: Convert workload into a starting hosting target

For a launch-stage WooCommerce and LMS site with 50 to 100 same-time active students during campaigns:

Resource area	Starting target	Reason
Hosting category	Managed WordPress plan with published dynamic limits, managed multi-site server with enough headroom, or VPS/cloud with managed operations	The site is dynamic and business-critical. Choose plans with clear dynamic capacity.
CPU	4 CPU-core equivalent as a practical starting target	Checkout, lessons, quizzes, reports, scans, and background jobs can overlap.
RAM	4 to 8 GB server or container RAM	PHP, database, object cache, background jobs, and system overhead need room.
PHP memory	512 MB minimum target, 768 MB to 1 GB for heavier stacks	WooCommerce plus LMS plus security plus builder and admin work can exceed simple-store assumptions.
PHP workers or same-time dynamic requests	Enough published capacity for checkout, logins, lessons, quizzes, and admin work during campaign traffic	Logged-in traffic and checkout cannot rely on full-page cache.
Object cache	Redis or equivalent strongly recommended	Reduces repeated database work for logged-in and dynamic pages.
Database	Clear database limits, slow-query visibility, backups, and upgrade path	LMS progress and WooCommerce orders create many reads and writes.
Storage	NVMe or strong SSD, file count limit checked, backup and staging storage understood	Course images, PDFs, logs, backups, and staging can grow quickly.
Email	SMTP or transactional email service	Protects order, enrollment, password, and course notifications.
Backups and staging	Daily backups minimum, easy restore, staging included	Course and commerce updates need safe testing.
Upgrade path	One-step resource increase or migration path before launch	Campaign spikes and course launches can outgrow the first plan.

Step 4: Choose the shortlist

Shortlist plans that meet these conditions:

Check	Required answer
Same category comparison	Compare managed dynamic WordPress plans with managed dynamic WordPress plans.
Source depth	Pricing page and technical docs checked.
Dynamic limits	CPU, RAM, PHP memory, PHP workers published or confirmed.
Object cache	Available and supported.
Database	Limits, backup, restore, and monitoring path clear.
Email	SMTP or transactional path available.
Support	Support scope includes WordPress hosting issues, logs, cache, backups, and restore.
Upgrade path	Resources can grow without rebuilding the project under pressure.

Starting Recommendation

For this project, recommended starting point:

- 4 CPU-core equivalent
- 4 to 8 GB RAM
- 512 MB PHP memory minimum, with 768 MB to 1 GB for heavy admin, import, or report work
- Object cache
- Strong SSD or NVMe storage
- Clear file count limit
- Staging
- Daily backups with restore
- SMTP or transactional email
- Published or support-confirmed same-time dynamic request capacity
- Upgrade path before launch

This is a practical launch baseline for a paid education commerce project. Adjust after measuring the real site.

What to Measure After Launch

Area	What to inspect
Checkout	Response time, failures, payment webhook timing, order email timing
Login and account pages	Response time under same-time users
Lessons and quizzes	Response time, progress writes, slow queries
Admin screens	Orders, reports, course editing, student lists
PHP memory	Memory errors, peak usage, fatal errors
CPU	Sustained CPU pressure during campaigns, imports, scans, reports
PHP workers or same-time requests	Queues, 502 or 504 errors, slow dynamic requests
Database	Slow queries, table growth, write activity
Cache	Public cache hit rate, object cache hit rate, cache exclusions
Background jobs	Cron delays, failed emails, stuck actions

Sample Recommendation

Use this as a model answer:

1. This is a WooCommerce plus LMS site, so the main risk is dynamic logged-in traffic, checkout, course progress writes, emails, and background jobs.
2. Compare managed dynamic WordPress plans or VPS/cloud plans with the same resource-isolation model.
3. Start around 4 CPU-core equivalent, 4 to 8 GB RAM, 512 MB PHP memory minimum, object cache, staging, strong SSD or NVMe storage, and transactional email.
4. The plan must publish or confirm CPU, RAM, PHP memory, same-time dynamic request capacity, database limits, file count limits, backups, and upgrade path.
5. Monitor checkout, logins, lessons, quizzes, admin reports, memory, CPU, PHP worker queues, cron, email, and database slow queries during test launch.
6. When the site shows memory, CPU, worker, or database pressure, upgrade resources before adding more courses, students, heavy extensions, or campaigns.

References

- WooCommerce server requirements: <https://woocommerce.com/document/server-requirements/>
- LearnDash technical requirements: <https://www.liquidweb.com/help-docs/software/learndash/references/requirements/>
- Defender requirements: <https://wordpress.org/plugins/defender-security/>
- SmartCrawl requirements: <https://wordpress.org/plugins/smartcrawl-seo/>
- Hummingbird requirements: <https://wordpress.org/plugins/hummingbird-performance/>
- WP Mail SMTP supported PHP versions: <https://wpmailsmtp.com/docs/supported-php-versions-for-wp-mail-smtp/>